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REGULAR PAPER

Genetic Algorithm-based Control of a Two-Wheeled Self-Balancing Robot

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Abstract

Mobile robots are becoming increasingly popular in a wide array of applications: industrial, item delivery, search and rescue, space, social, and entertainment. A two-wheeled self-balancing mobile robot is a statically unstable non-linear system with strong coupling dynamics. Common practices in the development of control systems for such robots are either to linearise the region of application to be used with linear controllers or to use complex nonlinear controllers such as fuzzy logic, sliding mode, and neural networks. However, self-balancing robots are still restricted by the travelling distance needed to regain an upright stance, the length of settling time, high overshoot and lack of resilience to external disturbance. In this paper, we are proposing a novel genetic algorithm-based switching control to evolve more effective control parameters and increase autonomy. Differently from previous work a genetic algorithm has been used to select the parameters in a sliding mode control and a switching-algorithm-based controller of a two-wheeled self-balancing mobile robot. The performance of the proposed controllers is assessed in simulations using the CoppeliaSim environment. The tests used dynamic criteria (distance travelled, maximum angular deviation), control criteria (settling time, % overshoot). The results showed that the genetic algorithm-based control has better performance in the 55 degree recovery, impulse response and variable inclination tests and that switching algorithm-based control shows better performance in step response tests. The results produced by the evolutionary algorithm are often able to perform better than their analytic counterparts. This shows the potential of meta-heuristic algorithms to obtain solutions for optimization problems encountered by statically unstable non-linear systems in unstructured and fast-changing environments.

Keywords TWSBR · CoppeliaSim · Genetic algorithms · Switching control · Sliding mode control · PID control

1 Introduction

Mobile robotics is a rapidly growing subset of robotics with widespread use in highly diverse environments.

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As a relatively new, growing field there is great potential for improvement in the existing technology as well as the creation of new methods, designs and techniques to tackle the complex challenges being faced [1].

Over the last decades the presence of mobile robots has become more commonplace in a multitude of applications: autonomous transportation across factory floors [2]), shelf stacking [3]), Segways' human transportation [4], door-to-door food delivery [5] and household cleaning [6]). Mobile robots can be either dynamically stable or statically stable (four-wheel robots). Dynamically stable mobile robots actively balance, as shown in humanoids (e.g. Atlas) and quadrupeds when they are on three legs (e.g. Spot, which, like Atlas, is a Boston Dynamics product). Compared to legged robots, wheeled robots are more energy efficient, easier to control and have a simpler mechanical structure. A subset of dynamically stable mobile robots are two-wheeled, self-balancing robots (TWSBRs), like Boston Dynamics' Handle robot [7], Tencent Robotics' Ollie [8] and Ascento [9] that dynamically balance to stay upright. Their mechanical structure comprises of two coaxial wheels at the base of a main body. These robots possess the inherent ability to re-orient themselves to maintain an upright position and rotate on the spot. This characteristic results in high manoeuvrability that enables operation in more cluttered environments and confined spaces compared to their three, four or multi-wheeled counterparts. Because two-wheeled robots lean forward and backwards to gain balance, they can lean into the incline to maintain stability. TWSBRs can also accelerate quickly without falling. Having only two wheels means potentially larger wheels, that afford them the ability to traverse rougher terrain [10]. However, their mechanical arrangement requires constant active control to maintain an upright stance. This challenge drove an intense research effort aimed at finding the best control algorithm to ensure dynamical stability. The complexity associated with TWSBR control makes it a suitable test platform for the development of control systems of automobiles, space crafts, military vehicles, and missiles [11].

Notwithstanding previous work in this field, self-balancing robots are still restricted by the travelling distance needed to regain an upright stance, the settling duration, percentage overshoot and resilience to external disturbance. This work presents a control approach that uses evolutionary algorithms to tackle these issues.

The evolutionary algorithm presented in this work aims to improve TWSBR behaviour by evolving more effective control parameters and increasing autonomy. Simplifying and generalizing optimization of control parameters through evolutionary computing processes allows for fast, unsupervised optimization. This facilitates quick adaption to changes in the environment and/or the various aspects of the TWSBR design. Differently from the state of the art, an evolutionary algorithm has been used to select the parameters in a sliding mode control and a switching-algorithm-based controller of a TWSBR (Fig. 1). The performances of these algorithms have then been compared.

2 Background

The TWSBR can be thought of as an inverted pendulum on wheels pivoting about the common wheel axis. By moving the wheels either towards or away from under the center of mass (CoM), the angular deviation and velocity of the TWSBR can be controlled. Hence, the motion of the system is coupled to the orientation of its body. The main objective in TWSBR control is to remain balanced and avoid falling down.

Various types of control exist that address the stability problems faced by TWSBRs. A popular approach is the combination of Proportional-integral-derivative (PID) and Linear-quadratic regulator (LQR) for the control [12–14]. LQR and PID were also used for input tracking and disturbance rejection [15] and for position control, upright (vertical) balance angle, and their disturbance rejection capabilities [11]. Experimental results of some of these studies used a physical TWSBR to show that self-balancing can be achieved close to the upright position [13, 14]. LQR was also coupled with neural networks for TWSBR control [16], specifically, LQR's parameters are utilised to initialise the neural network. Unfortunately, the linear controllers only achieve good performance in small tilt angles and lack in robustness in dealing with TWSBR uncertainties [17]. Sliding Mode Control (SMC) is very suitable for non-linear problems in state and time. It is often used in robotics where the environment is variable and unpredictable and it is naturally fit for use in TWSBRs [18]. However, SMC is affected by the chattering phenomenon of the sliding surface [17].

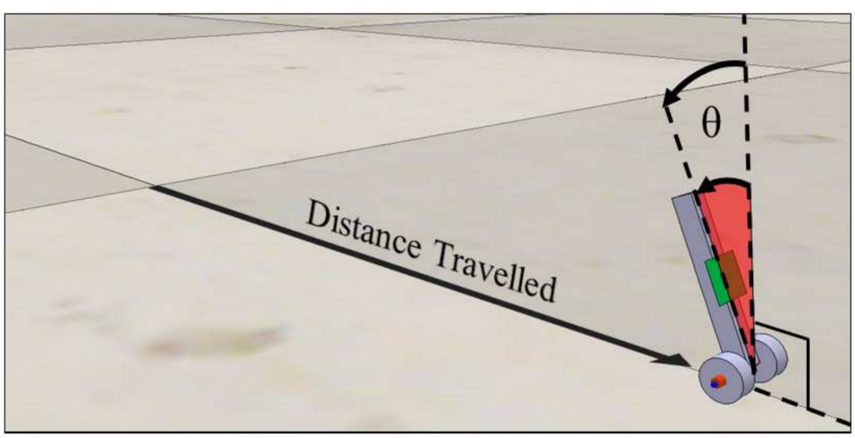


Fig. 1 The TWSBR in the CoppeliaSim environ. The distance travelled (d) and the angle of inclination (θ) are metrics used to evaluate the TWSBR behaviour and performance. θ is the angle of the TWSBR body from vertical. If θ is zero, the TWSBR is completely upright

Artificial intelligence can approximate non-linear systems. For this reason, artificial intelligence integrated to conventional controllers is a good strategy to overcome these control issues. Artificial neural networks are a popular adaptive control strategy for TWSBRs [19] and have been used to online estimate the unknown model parameters SMC control of TWSBRs [17]. Fuzzy control is also a strategy often utilised for non-linear systems. Fuzzy logic control of TWSBRs was proven by simulation results to result in successful self-balance [20]. Fuzzy controllers based on a Takagi-Sugeno model were designed, showing a fuzzy model-based approach to designing the controller [21, 22]. Compared to linear controllers, fuzzy controllers add robustness to uncertainties deriving from external disturbances [23]. In general, the common point of these studies is applying the self-learning capabilities of the neural networks and fuzzy logic to deal with nonlinearities and uncertainties in the control process of the TWSBR [17].

Evolutionary algorithms (EA) are a subset of bioinspired, meta-heuristic computational algorithms. EAs utilize a multitude of individuals, across successive generations to converge to a desired solution through simulated evolution. They can be used to efficiently explore a solution space to find a solution specifically adapted to and shaped by its environment. Genetic algorithms have been used to optimise the parameters of PID controllers in manipulators [24, 25]. Genetic algorithms are very appealing for tuning PID parameters [26] and consequently have been used in multiple studies [27–29]. A study focusing on mobile robots found that evolutionary algorithms are an effective tool for solving optimization problems in control [30]. Although a study compared genetic algorithms unfavourable compared to neural networks for TWSBR control [31] and EA may become stuck in local maximuma indefinitely, an evolutionary algorithm is successfully used to find solutions of fuzzy input and output membership functions and to determine a rule base in [32]. In addition, a genetic algorithm was used to tune the PD and PID controllers parameters of a TWSBR [33] and a genetic algorithm is used for the optimisation of a self-balancing robot in [34]. Furthermore, genetic algorithm and bacteria foraging optimization algorithm are used to obtain values for controller parameters in a Segway controlled via the LQR technique [35]. Given this evidence of success, an evolutionary algorithm was chosen for our work.

Meta-heuristic algorithms are most often used when traditional methods are either too slow or cannot obtain an exact/analytical solution. Their benefits become apparent when the solution space is too large, or computation power and time are limited [36]. This is the case with TWSBRs and the second reason that an EA was chosen to select control parameters. The practicality of the meta-heuristic algorithm can be improved through its combination with simulations, allowing the real-life uncertainty to be integrated within the model. Juan et al. [37] show that the benefits of meta-heuristics coupled with simulations (simheuristics) far outweigh any downsides and should be considered as a convenient first resort given that simulations are readily available and affordable [38].

Notwithstanding the multiple examples of the use of EAs in TWSBR control, it has not, so far, been used to refine SMC and TWSBR switching control algorithms. Switching control is a control mode for TWSBRs proposed by Murasovs et al. [39] which adjusts its control strategy based on the state of the system, employing different control algorithms depending on the deviation angle of the TWSBR. This allows each control algorithm to operate where it performs best, switching to another control type when advantageous. These two control types are defined by parameters which have been obtained analytically. In this work, pure switching control will be compared to switching control enhanced by evolutionary algorithms.

In order to ensure that an accurate robot model is used and that the conducted experiments are comparable with earlier work, the dynamic model of the TWSBR has been based on a physical robot used in [39] and [40]. Its physical parameters are described on Table 1.

2.1 Control Techniques

In the next three sections we will describe the three approaches that were used in our system and their use in previous studies. Firstly, the TWSBR dynamic model used in this work is described. Secondly, the switching algorithm is detailed. Thirdly, the EA approach we use is discussed.

2.1.1 Dynamic Model of TSWBR

Murasovs et al. [39] use the procedure outlined by Dai et al. [18] to define the SMC surface for the desired TWSBR behaviour. Following this process, the state-space model of the TWSBR can be written as

$$M(X)\dot{X} = F(X) + U, \tag{1}$$

where matrices $M(X)$, $F(X)$ depend on X and the Lagrangian dynamics of the TWSBR, while matrix U depends on the wheel torque. X is the matrix which fully describes all relevant parameters of the current physical state, given by

$$X = \begin{bmatrix} \dot{x} \\ \theta \\ \dot{\theta} \end{bmatrix}, \tag{2}$$

where $\dot{x}$ is the straight line velocity, θ is the angle of the TWSBR body from vertical and $\dot{\theta}$ is the rate of change of this angle. The error of the system e is given by the difference

Table 1 Physical parameters of the TWSBR model [40] used for state-space model and simulation parameters

Parameter	Value	Description
M_b	0.595 kg	Mass of robot's body
M_w	0.031 kg	Mass of robot's single wheel
J_b	0.0015 kgm^2	Body's moment of inertia about CoM
J_w	5.96e−05 kgm^2	Wheel's moment of inertia
r	0.04 m	Radius of wheels
L	0.08 m	Distance from wheel axle to robot's CoM
K_e	0.468 Vs/rad	EMF constant
K_m	0.317 Nm/A	Motor constant
R	6.69 Ohm	Motor resistance
b	0.002 Nms/rad	Viscous friction constant
g	9.81 m/s^2	Gravitational constant

between the desired X_d and current states. In this case, this is a 1-by-3 matrix:

$$e = X_d - X = \begin{bmatrix} x_d \\ \theta_d \\ \dot{\theta}_d \end{bmatrix} - \begin{bmatrix} x \\ \theta \\ \dot{\theta} \end{bmatrix}. \quad (3)$$

The cross-product of the error matrix e with C is then used to obtain the sliding surface s as

$$s = Ce, \quad (4)$$

and

$$\dot{s} = -\epsilon * sign(s) \quad (5)$$

where C is a 3-by-1 matrix whose parameters are to be determined:

$$C = \begin{bmatrix} c_1 \; c_2 \; c_3 \end{bmatrix}. \quad (6)$$

The original selection for the parameters c_1, c_2, c_3, and ϵ was obtained through an iterative process [39]. Most of the potential solutions cause instability, and the chosen values provide a reponse with a settling time quicker than that of PID. In this work, the values are obtained through an EA to improve the settling time and avoid local minimum. The three components of C manifest themselves as the genes in the EA which will evolve over generations to control the behaviour of the TWSBR as specified by the EA design.

2.1.2 Switching Control

The switching control algorithm applied to the TWSBR utilizes a combination of SMC and PID control for the TWSBR, switching between each mode depending on the angle of inclination of the robot body. This combination has shown significant performance benefits over each control mode on its own as each control type takes over in the regions where it performs best. In this case, PID takes over in smaller angles where the control problem is more linear [39], and SMC at greater angles where there is greater non-linearity. In [39] the switching angle is set to ±5 degrees from vertical (see Fig. 2), but further work is necessary to assess this choice as other combinations of PID gains, SMC surfaces and switching angles may produce an even better response. The switching angle, along with the SMC values is an additional parameter that can be addressed by the EA.

2.1.3 Evolutionary Algorithms

Evolutionary algorithms model the process of natural selection in a given population, where the characteristics and/or behaviour of the individual members in successive generations are altered through genes (i.e. parameters) such that the 'fittest' individuals' characteristics prosper and the least so 'die off'. The resulting individual's characteristics converge towards the solution over the span of multiple generations thus, providing a solution optimized for the given problem [41].

The fundamental EA cycle [36] consists of the creation of an initial generation, usually with randomly generated parameters followed by the cycle of four steps (described below) [41] to evaluate and populate the next generation until the optimal condition is met.

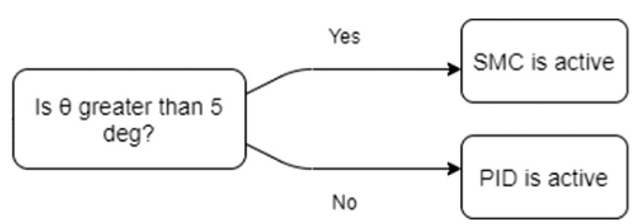


Fig. 2 The fundamental principle embodied in switching control presented in [39]

After the first generation is populated with individuals with random parameters, the following four steps are repeated until the end of the EA process:

Step 1 - Fitness (Evaluation) Function. The objective of EA is to assess how well the individuals in the population perform. This is necessary to minimise problematic performance characteristics and maximise beneficial ones. Overall, a fitness (or evaluation) function is defined to include all metrics relevant to the performance of the TWSBR [41].

Step 2 - Selection. The objective of this step is to distinguish individuals based on their quality, and in particular, to allow better ones to pass on their genes to the next generation. Such individuals have the highest fitness score relative to others in the population. This approach is very probabilistic, where high-quality individuals have a high chance to be selected for the next generation, and low-quality individuals have a small chance; otherwise, EA might get stuck in a local optimum [41].

Step 3 - Mutation and Crossover This step is needed to create new individuals from the ones selected from the previous generation. By altering represented parameters, the mutation of the individuals generates a modified population to be tested in the next generation. This process is always stochastic, as its output depends on the outcomes of a series of random choices [42]. This is followed by a process named crossover (or recombination), where the information selected from two individuals is merged to produce two or more offspring individuals. This process is also stochastic [42].

Step 4 - New generation The new generation may be comprised of completely new individuals, replace only the worst-performing individuals or even maintain just the fittest (elitism). In this work, each new individual is a result of the combination of processes outlined above, with no individuals kept from previous generations. This choice is made to encourage a faster convergence through the increased number (and therefore variety) of different simulated individuals.

2.2 Contribution to the State of the Art

In this work, for the first time, an EA has been used to select the parameters in a SMC and a switching-algorithm-based controller of a TWSBR. In addition, for the first time, the performance of an EA-enhanced SMC control has been compared to the performance of an EA-enhanced switching algorithm control in a TWSBR.

One area in particular which needs further exploration is the TWSBR behaviour on uneven, rough terrain as most of the literature presents control designed for horizontal, flat terrain [43]. The ability to effectively traverse uneven terrain would give TWSBR the ability to work in more unstructured environments, more similar to real-life conditions. Although the TWSBR control from Murasovs et al. [39] is tested on a smooth 20° ramp, no analysis is done on the behaviour when the surface gradient changes faster than the algorithm can settle, or when the surface is not perfectly smooth. Such surfaces will be considered in this work, as a small variation in the contact angle between the wheel and terrain produces significant disturbances [44].

Furthermore, a limitation of previous work that uses a switching algorithm is the assumption of perfect friction with the ground [39]. Frictions within the motors and slip between the wheels and the ground contribute to a highly non-linear error that negatively impacts the performance of control algorithms [45, 46]. Consequently, quantifying the effectiveness of the proposed algorithms under variable slip conditions is worthwhile.

In this study, seven different experimental setups are used to test and compare four different control systems for TWSBR: SMC, SMC with evolutionary algorithm (ESMC), the switching algorithm (SW) and the switching algorithm with evolutionary algorithm (ESW). The methods used in the six experiments for the comparison and elucidated in the following section. The results of these experiments are detailed in Sect. 6 and discussed in Sect. 5.

3 Materials and Methods

3.1 Simulation Environment

CoppeliaSim [47] was chosen as the simulation environment to ensure consistency between previous work and the new simulations. In addition, CoppeliaSim has the advantage of various options for programming functionality, including scripts attached to robots and separate programs that connect to it via the RemoteAPI. [48]. In this study CoppeliaSim was connected to MATLAB, where the EA was implemented. The physics engine Vortex [49] was utilised as part of the CoppeliaSim setup. Despite using simulation software, the experiments were performed in real time, and using real-world mechanical parameters to mimic the realistic conditions. DC motors have actuation frequency of 250 Hz which equates to 0.004 s. This is the time step for the simulation, hence the processing and sampling time are both equal to that. The state of TWSBR is obtained through an accelerometer (for θ) and gyroscope (for $\dot{\theta}$) and in the main chassis, and the speed of the TWSBR is obtained from the information on the wheels' angular speed and the wheel circumference [50]. This information is used in the feedback loop of the control system to adjust the dynamics of TWSBR.

3.2 EA Structure

The chosen EA for the following experiments is structured based on the principles in [41]. The following section outlines the EA implementation, starting from its initialisation, detailing the fitness function metrics used and explaining the end conditions utilised.

3.2.1 EA Initialization

Before the EA algorithm can run, the parameter-space, population size, fitness function and other fixed variables relating to the EA need to be defined. The parameter-space is the range of all possible values that each of the optimization variables can take. The parameter-space for this control problem it is the three-dimensional domain of the product of the range of SMC parameters or four-dimensional domain of the product of the range of SMC parameters and switching angle values.

The first generation is populated with random decimal values within the defined parameter-space. This gene representation method was selected over traditional binary representation as typically the gene values reflect something about the problem [51]. The genes of the EA define the switching angle of the switching algorithm and the sliding surface of the SMC which then used control the voltage output to the wheels depending on the state of the TWSBR. For the pure SMC EA, the matrix parameters (Eq. 6) are limited between ± 1 with a resolution limit of 4 decimal places, producing a state space of $2 \times 10^{4^3}$ possible gene combinations. The switching EA multiplies this by 75×10^4 as the angle parameter space is included.

The generation size and number of generations significantly affect how quickly the robots will converge to a solution as well as how long the simheuristic process will last [38]. Larger generation sizes are able to converge in fewer generations and are less susceptible to getting stuck in local maxima due the presence of more individuals and therefore more variety, but this significantly increases the total simheuristic computing time.

In our case it quickly became apparent that better results could be obtained from larger sets of 25 generations with 80-100 individuals each.

Finally theding fitness function is defined starting with the metrics to be used.

The metrics with which the TWSBR's performance is measured are defined below:

1. **Settling time** - the time it takes for the TWSBR to reach and remain within a small envelope (2%) of the upright standing position after any disturbance to its state. For the TWSBR, the settling time is used to assess how fast both the velocity and the angular deviation reach their desired state.
2. **Maximum Overshoot** - the peak value seen in the response of the TWSBR as it exceeds the desired target state. The target value can be either a specific angle or velocity value, depending on the experiment. Lower overshoot values are better, demonstrating to what extent the control algorithm is over/under-damped and how the TWSBR tends to its final state.
3. **Distance travelled** - the displacement from the origin (d) of the TWSBR as it recovers from an external disturbance. This metric is crucial in ensuring the robot can recover in confined environments and can fully stabilize itself quickly. It is applied only to experiments where the desired state is static (e.g. angle recovery). Figure 1 shows the physical representation of this value in the simulation.
4. **Cumulative angular deviation** - the sum of the magnitude of the angular deviation from the upright position during the simulation, essentially keeping track of how upright the TWSBR is throughout its response. The value of θ in Fig. 1 shows this angle, whose integral with respect to time gives the cumulative angular deviation. In addition, it is helpful in implicitly reducing jitter. and power consumption as fewer and smaller adjustments are needed when closer to equilibrium.
5. **Power Consumption** - This is the total power in Watts used by the two TWSBR motors during an experiment. It is obtained from the voltage and current inputs to the motors.

3.2.2 Simheuristic Process

The simulation time (ranging from 3-10 s depending on the experiment) provides sufficient time for even the slowest robots to stabilize, and enough time for the unstable robots to fail in doing so. Once all of the individuals in a generation have been simulated, selection takes place. Their performance metrics (upright time, maximum deviation, distance travelled, etc.) are used to calculate a fitness value for each individual. The fitness values are squared to bias the selection towards larger values and then normalised. Finally, a random, weighted 'roulette wheel' algorithm picks two parents with a selection probability proportional to their fitness to produce one new individual.

3.2.3 Switching EA and SMC EA

The EA for switching was applied in conjunction with SMC optimization in an attempt to reduce power consumption and improve the transient response of the system. Designing an EA to optimize the SMC requires a three-dimensional search-space, one dimension for each of the SMC parameters in Eq. 6. The term ESMC is used to refer to the SMC surface obtained through the evolutionary algorithm. Since the early SMC generations have very poor performance while PID does not, the EA rapidly takes advantage of that by preventing SMC from ever taking over in the switching control. The already optimized PID could take over and perform better than any of the initial SMC surfaces. This is unacceptable as it prevents any meaningful evolution of SMC values. For this reason a constraint was placed, only allowing the switching angle optimization to take place after a certain amount of generations (commonly 80% of maximum), ensuring the SMC control becomes functional in that time. This type of control will be denoted as ESW (evolutionary switching) moving forwards.

3.2.4 End Condition

There is a variety of ways to determine when to end an evolutionary algorithm. The end condition can be (but not limited to) when a certain goal or target is met, when a given number of generations elapse, or when there is a natural stagnation in the EA improvement [41]. All three end conditions were utilized initially, but the target and stagnation conditions were not used as they could be 'tricked' by anomalous individuals or other unexpected situations. Examples of the target end condition failing include terminating a simulation when an individual marginally meets the required metrics while severely underperforming in all others. The stagnation end condition is occasionally triggered when a brief period of minimal change follows a period of rapid improvement into a local maximum (a common phenomenon) triggering the termination condition with poor results. Consequently, the fixed generation limit was chosen through trial and error to ensure enough time was provided to veritably reach a global maximum.

3.3 Simulated Experiments Setup

The first four experiments have been designed to test a specific response characteristic of the robot and for a general assessment of the robot's performance. In addition, more complex tests have been conducted, where the relevant strengths and weaknesses of the analytical and evolutionary parameters can be compared. These include a surface with abrupt and frequent gradient changes along its length, a rough surface, and experiments with reduced friction.

3.3.1 Angle Recovery

This test sets the robot body at an angular offset to upright at the start of the simulation, assessing how quickly (if at all) the robot is able to recover itself, how quickly it can settle back to its upright stance, and how far it has to travel to achieve this. The evolution simulation was carried out with 100 individuals for 25 generations. In this test, the EA attempts to evolve a new sliding surface for the TWSBR. ESMC control that is able to withstand and recover from an initial deviation of 55°, allowing for a direct comparison with the previous work [39]. As well as the above, the EA simultaneously optimizes for the switching angle of the ESW algorithm. The simulation time provided was three seconds. The fitness function for this experiment was

$$\text{Fitness} = \frac{50t_u}{E(100\theta_{max} + \theta_{tot} + u_{max} + d)}, \tag{7}$$

where t_u is the total upright time, θ_{max} is the maximum overshoot, θ_{tot} is the cumulative angle of inclination, u_{max} is the maximum velocity and d is the distance travelled during the performance. The focus of the angle recovery tests is to reduce the power consumption E as much as possible while simultaneously optimizing the physical performance. The metrics that must be increased for better performance are found in the numerator, along with an associated weight value which affects the impact that specific metric has on the fitness value relative to other metrics. The metrics that must decrease for better performance and their associated weight values are found in the denominator, resulting in an increase in the fitness value as they decrease.

For the experiments such as this and the impulse recovery, where a still, upright position is the target, the settling time of the velocity and the overshoot of the angle of inclination are the most important metrics as they indicate the time taken to become still and how upright the TWSBR remained respectively. The velocity and angle overshoot values are inversely proportional, naturally, as the angle overshoot decreases as the velocity increases and vice versa.

3.3.2 Impulse Recovery

In Murasovs et al. [39] this test features a pendulum of a large solid block that swings and hits the top end of the TWSBR body while in the fully upright position. This is modelled as a perfectly plastic collision, exerting an instantaneous horizontal force whose magnitude is dependent on the mass and length of the swinging block. This test was slightly modified to simulate a scenario closer to real-world dynamics. In this new test, we apply a force to the top end of the robot body using a built-in CoppeliaSim command. This enables accurate measurement of the resistance against external disturbances. For this test, the ability of the robot to withstand the largest external force and recover from the largest angle of inclination was used to examine fitness. Hence, the optimal EA fitness function for this test is:

$$\text{Fitness} = \frac{50t_u}{E(\theta_{max} + \theta_{tot} + 10u_{max} + 1000d)}. \tag{8}$$

t_u is the total upright time, θ_{max} is the maximum overshoot, θ_{tot} is the cumulative angle of inclination, d is the distance travelled, E and u_{max} are the total power consumed and the velocity overshoot of the robot respectively.

The maximum force tolerated by the original SMC and switching algorithms was found before the evolutionary simulation was run. This was used as a reference starting point for the EA. Multiple evolutionary simulations were run with the external force increasing in magnitude in increments of 5 N. This was repeated until none of the 100 individuals in a generation was able to successfully arrive at a solution after ten generations.

3.3.3 Step Response

The classic step response test assesses the overshoot and settling time of a system, much like the recovery angle deviation test. In this case, a step response is applied to the desired state of the velocity of the robot rather than the physical velocity. To meet this target, the TWSBR must first accelerate in the opposite direction, to offset its center of mass towards the right direction, and then regulate this offset to meet the required speed. The step response evolution simulation was carried out with 100 individuals for 20 generations. This experiment changes the target velocity state of the robot from still to $1ms^{-1}$ As the robot tries to reach the velocity step profile, the overshoot and the settling time are used to evaluate its fitness:

$$\text{Fitness} = \frac{100t_u}{\theta_{max} + \theta_{tot} + u_{max} + 0.1d}. \tag{9}$$

The TWSBR that is able to accurately meet the velocity in the shortest time possible and with the lowest overshoot is considered the fittest in this experiment. For the step response tests, consistency in the velocity and angular deviation is the goal, so their overshoot and settling times are equally important.

3.3.4 Ramp

In this experiment, the TWSBR is made to traverse an incline and decline while trying to maintain a constant velocity.

The TWSBR in [39] has only been tested on ramps of ±20°, (see Fig. 3). Consequently, the EA individuals' evolution is tested on the same ramp with flat but rough terrain. Attempting to evolve a control algorithm to tackle inclines requires staying upright and moving forwards first. Instead of trying to evolve from scratch, the control algorithms evolved in the step response test (possessing the ability to move forwards) will be used for this experiment.

3.3.5 Variable Inclination

Furthering the ramp experiments, a series of varying inclines and declines ranging from ±30° to horizontal are traversed by the TWSBR in this test (see Fig. 4).

Each change in the variable surface inclination requires a small adjustment in the velocity and angle of the TWSBR. The variable surface experiment tests if the robot can reliably adjust and remain upright when the surface profile changes faster than the response can settle.

3.3.6 Rough Terrain

The TWSBR is also tested on a surface with small irregularities comparable to the diameter of the wheel (rough terrain, Fig. 5). The rough surface is designed to alter the contact point of the wheel and the floor to simulate a surface roughness with features of ±2.5mm, or 12.5% of the wheel radius.

Much like the step response and ramp experiments, the aim of the control algorithms in these varied surfaces will be primarily to maintain a constant velocity of 1ms^{-1} and an angle as close to zero as possible. If all else is equal, metrics of lower importance such as cumulative angle and power consumption will be used to evaluate their respective performance.

3.3.7 Reduced Friction

All of the above tests assume perfect contact between the TWSBR wheels and the ground, and no slip. Often this is not the case, especially in industrial and commercial environments involving dust or liquid spills.

Reduced friction means the wheels must spin faster to produce the same velocity difference as before. Due to the 45 V limit on the motors (and thus on the maximum rotational speed of the wheels) a reduction in the contact friction naturally reduces the maximum angle that can be tolerated before the state is irrecoverable. Obtaining the minimum friction that

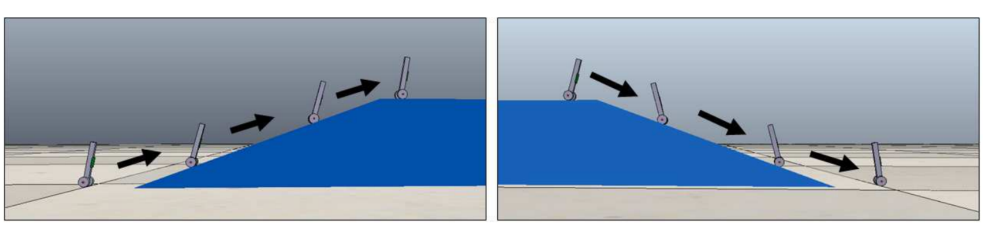


Fig. 3 The 20° incline (left) and decline (right) used in the ramp experiment. From [39]

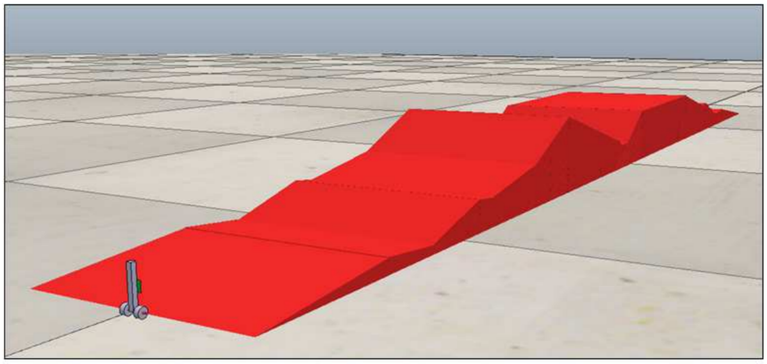


Fig. 4 The variable inclination surface traversed by the TWSBR in CoppeliaSim

can be tolerated before the 55° deviation is irrecoverable, will indicate the robustness of each algorithm in such conditions. Similarly to the step response, an initial angle deviation is required to shift the TWSBR into forward motion. However, as the angle and consequently speed increase, faster wheel rotation is required to recover.

The physics engine in CoppeliaSim uses Coulomb's law of friction to quantify the coefficient of friction (CoF) between two surfaces (μ) as a number between 0 and 1, given by

$$F_t = \mu F_n, \tag{10}$$

where F_t is a force tangent and F_n is the force normal to the contact plane between the wheel and ground. As long as F_t is less than μF_n slipping does not occur [52]. The normal force is given by $M_b g$ where M_b is the mass of the robot (0.595 kg) and g is the acceleration due to gravity (9.81ms^{-2}). Therefore, a maximum CoF of 1 means that slip will occur if F_t exceeds 5.94 N, decreasing proportionately as μ decreases.

The angle recovery and step response tests are repeated with each control system, reducing the wheels' CoF until they topple. In this way, the minimum friction that can be tolerated is determined, with lower friction values indicating more robust control.

4 Results of the Simulated Experiments

This section details the results of the seven experiments conducted to test the effectiveness of adding EA to the SMC and switching algorithm based control.

4.1 Angle Recovery Results

The behaviour of the four different control systems at angles of 55° can be seen in Fig. 6. Immediately, a notable observation is that all four control algorithms have comparable physical responses, whereas the voltage responses vary more significantly.

The multi-objective benefit of the EA reveals itself in the 55° test, where there are notable simultaneous performance improvements in multiple metrics. The primary metric based on the fitness function is the angle overshoot, followed by the velocity overshoot performance. In the angular deviation profiles (see a first trough in the negative quadrant of the angular deviation in Fig. 6), the SMC and switching algorithms peak at a maximum of 28.8° (0.5 radians) the ESMC has an improved maximum overshoot of 27.9° (0.48 radians) and the ESW a worsened overshoot of 29.8° (0.52 radians).

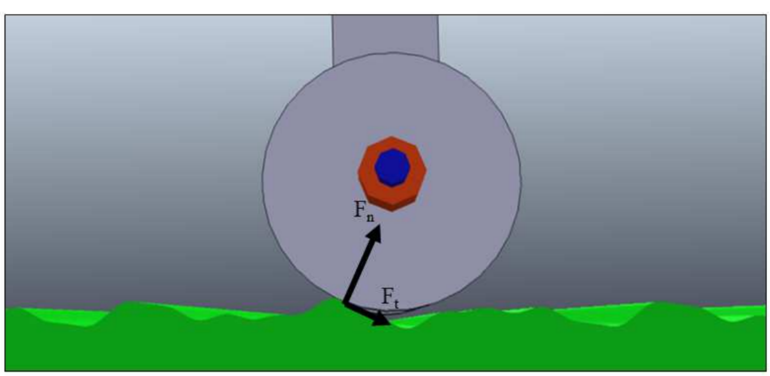


Fig. 5 Close-up view of the surface features of the rough terrain experiment. F_n and F_t show how the direction of the normal and tangential forces respectively change on this surface

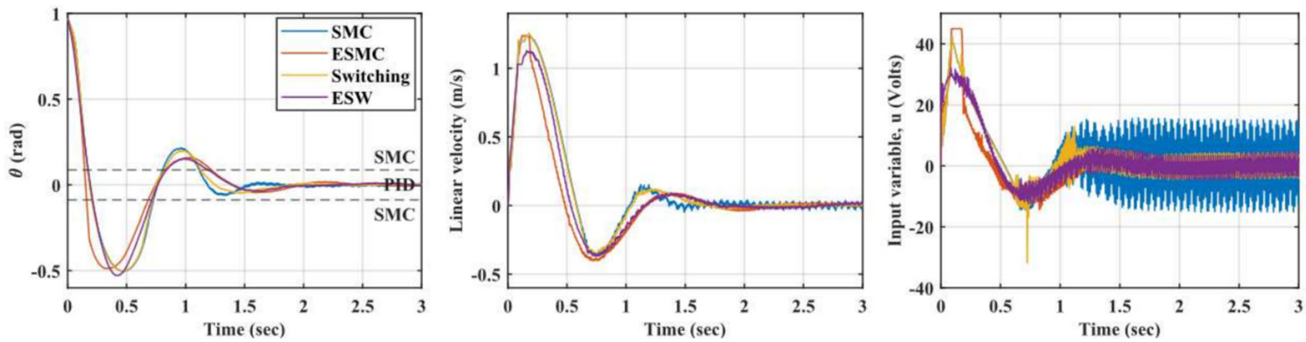


Fig. 6 TWSBR velocity, angular deviation, and motor voltage profiles in response to 55° angular offset

This last result is to be expected due to ESW reduced velocity overshoot. As in the Murasovs et al. work [39], the settling time of the velocity is faster for the SMC at 1.48 s compared to the 1.68 s of switching control. The evolutionary algorithms are of course not able to improve since this is not a factor used in this fitness function. The ESMC has a settling time of 1.87 s and the ESW algorithm has a settling time of 1.77 s respectively.

Whereas the SMC, ESMC and the original switching algorithms have a velocity peak of $1.24 \pm 0.01\text{ms}^{-1}$ the ESW value peaks at 1.13ms^{-1}, a 9.6% reduction. It must be noted, that the settling times are all functionally identical, though strictly speaking the pure SMC never settles as its velocity jitter regularly peaks up to 3.6%. The kinetic energy is shown in Fig. 7.

A critical benefit of the evolutionary algorithms (ESMC and ESW) is their capacity to achieve the above performance while travelling a shorter distance, a crucial factor if their use is to be considered in cluttered environments. Table 2 shows

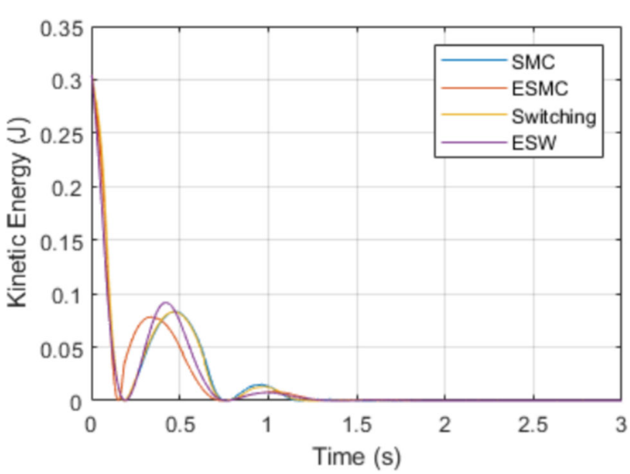


Fig. 7 Kinetic energy of TWBSBR in response to 55 ° angular offset

the distance travelled by each control mode during the angle recovery experiments.

In the 55° recovery the evolutionary algorithms are able to recover in significantly shorter distances than the analytical algorithms.

The voltage profile of the ESMC is much improved over the analytical SMC, with lower jitter and lower power consumption in the same three second window for both angle recovery tests. The ESW had a smaller improvement over the switching control, but nonetheless the power consumption and jitter remained low. Table 2 shows the exact values for each control system and each test.

These results indicate that the EA produces better results in the 55° test since more of the primary metrics and more overall metrics improved. The reason for this is difficult to quantify accurately, but given the existence of limitations and assumptions of the analytical models, it is probable that the meta-heuristic algorithm encountered a more optimal solution with regard to the specified fitness parameters.

In these experiments, the velocity and angle overshoots did not improve as much as would be expected. Conversely, metrics such as the power and distance travelled did, indicating either that they are further from the optimal point compared to the voltage/angle overshoot, or that their weight values were too large relative to the other metrics.

4.2 Impulse Recovery Results

In Murasovs et al. the PID was the worst performing control type, managing to withstand 615 N compared to the significantly higher SMC and switching control values: at 1260 N and 1450 N respectively. Figure 8 shows the maximum force withstood by each control system with the new external disturbance test. Since the values for the top three were very close each test was repeated ten times to ensure the outcome was not subject to errors within CoppeliaSim, the

Table 2 Comparison of main performance characteristics in angle recovery and step response experiments

Test	Criterion	SMC	ESMC	Switching	ESW
Recovery (55 deg) - θ response	Error band (rad)	0.0211	0.0189	0.0097	0.0124
	Distance travelled (m)	0.38	0.19	0.38	0.28
	Power consumed (W)	30.51	17.10	21.80	17.79
	Settling time (s)	1.53	1.87	1.75	1.80
Step response - $\dot{x}$	Distance travelled (m)	2.9453	2.003	1.971	0
	Power consumed (W)	96.1	92.1	47.8	47.7
	Settling time (s)	DNS	1.420	1.196	1.484
	% Vel. Overshoot	199.0	0.5	7.9	16.0
	% Angle Overshoot	101.0	61.5	45.4	22.8

The best results are in green, worst ones are in orange

values show the average value withstood and the maximum deviation of the ten tests.

As expected in this modified experiment, the maximum force values are considerably lower than the original mass and pendulum setup. In this experiment the ESW control outperforms all the other modes, withstanding up to 620 N consistently and averaging 621.5 N from the ten tests. The original switching and ESMC control are very closely matched at 619 N and 615 N respectively. Finally, the original SMC was able to withstand up to 567 N on average before toppling, significantly lower than the other three algorithms.

In the original external disturbance test the difference between the SMC and switching algorithms was 190 N, representing a 13.1% difference in their values. In the new experiment, the difference was 52 N or 8.4% difference. The ratio of the differences relative to the maximum value in the two algorithms is comparable, indicating the two different experimental designs are consistent with each other.

4.3 Step Response Results

Fig. 9 shows the output response of the control algorithms subjected to the a step response in the velocity. The response of the analytical SMC and switching algorithms is identical to the Murasovs et al. work [39], though the simulation duration is slightly longer.

From this, it is evident that the ESMC algorithm is initially able to meet the velocity step output more accurately than the original SMC algorithm. It greatly improves over the analytical SMC (which has 199% overshoot) with a practically identical settling time, but no overshoot. Problematically, it is not able to return to zero on the downward step. Due to the fitness emphasis on remaining upright and lower overshoot for the initial velocity step, the ESMC has little selective pressure to return back to zero. Different fitness functions and approaches were not able to overcome this effectively. The ESW algorithm improves on the ESMC algorithm, managing the downward step more effectively, but is not better than the original switching algorithm in its velocity profile. It has a settling time very similar to the original switching, but a marginally higher overshoot compared to the analytical switching.

Interestingly, both evolutionary algorithms have a considerably better angle response than the respective analytical solution, with the ESMC having an angle overshoot of 35.6° compared to 58.1° for the analytical SMC surface (39% improvement), and the ESW having an angle overshoot of 13.5° compared to 26.9° for the original switching algorithm (50% improvement).

When considering the voltage graph, the ESMC has reduced voltage oscillations (less jittering - smoother response) making it more comparable to the analytical switching control. The ESW performs similarly to the switching algorithm

Fig. 8 The maximum force the TWSBR can withstand without toppling over, based on the control algorithm used

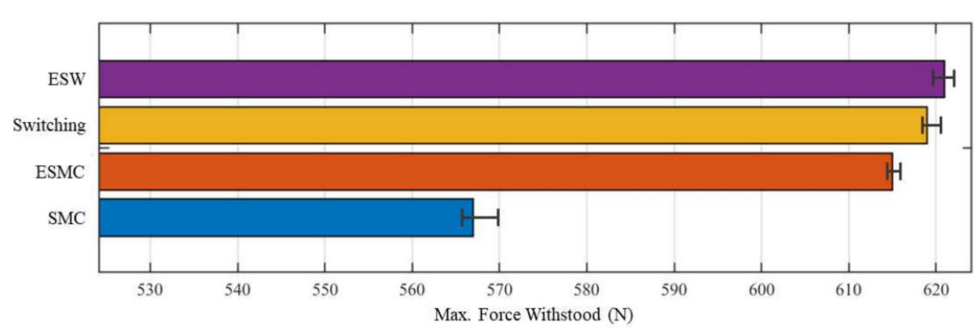
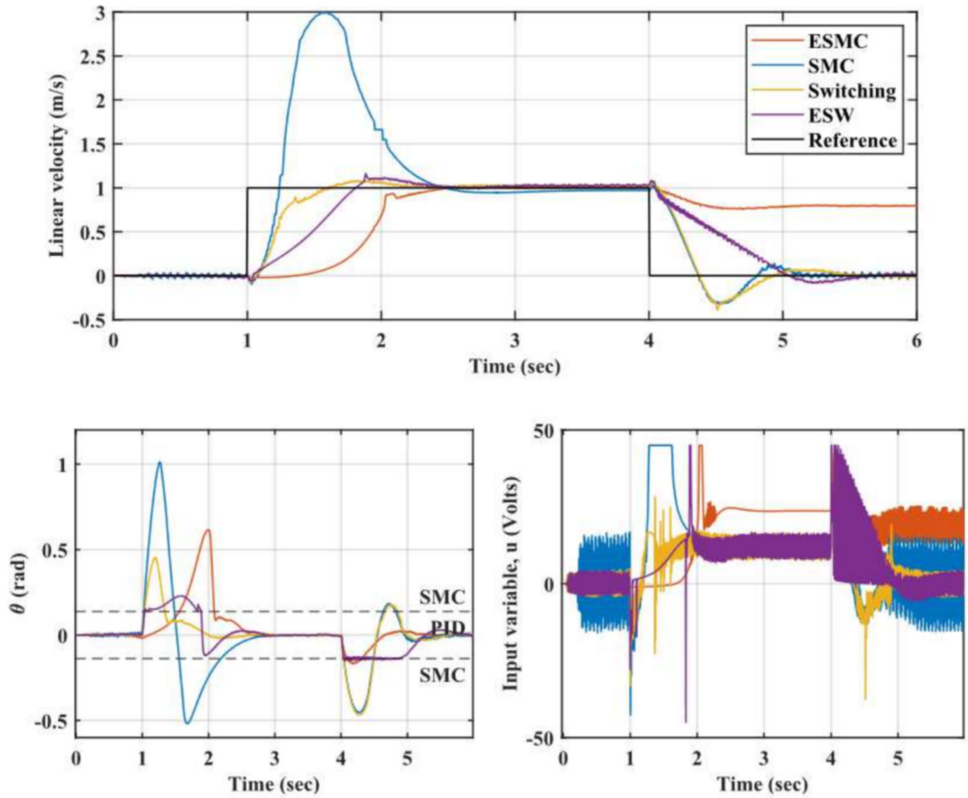


Fig. 9 TWSBR velocity profile in response to a unit step input (Reference Velocity)



other than a large feature on the negative step as the PID control 'fights' against the SMC to bring the steady-state error to zero. Finally, the power consumption differences marginally favour the evolutionary algorithms (see Table 2). The analytical SMC and switching algorithms consumed 96.07 W and 47.83 W respectively whereas the evolutionary algorithms consumed 92.14 W for the ESMC and 47.78 W for the ESW. The ESMC surface was able to use 4% less power than its analytical counterpart while the ESW barely improved at 0.1%. In practice these differences are small and may be attributed to simulation uncertainties, but the improvement increases exponentially with scale. The kinetic energy is shown in Fig. 10.

The ability of the ESMC to meet the exact speed without any overshoot is very promising, however it is overshadowed by the fact that the TWSBR does not stop afterwards. A way around this issue would be to use the current ESMC surface when stepping from zero to one, and a different surface optimized on the downward step when stopping/ reversing. The additional logic required to select a surface in real time based on the desired goal of the TWSBR would be straight forward to implement if optimal performance is key. With regards to the fitness function, the ESW algorithm performs much better than the switching control due to its matching velocity profile and significantly reduced overshoot and improved settling times in the angular deviation profile. However, the voltage oscillations present in the down step indicate that additional optimization is necessary; this was not possible in this work, due to to the limited computing resources available.

Table 3 shows the evolved parameters for the above control problems. Note the similarity in the first two tests, likely a result of the delayed inclusion of the switching angle metric making further evolution difficult or disadvantageous.

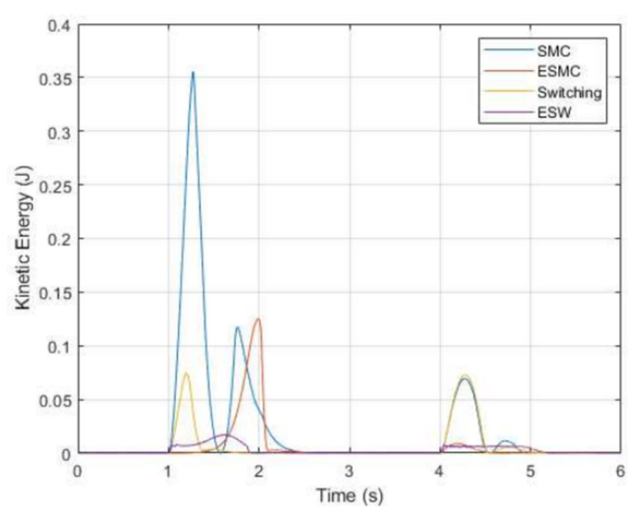


Fig. 10 TWSBR kinetic energy in response to a unit step input (Reference Velocity)

Table 3 The evolved control parameters for all tests

Test	Control Mode	c_1	c_2	c_3	Switching Angle [deg]
55° Recovery	ESMC	−0.6806	0.8897	0.3519	-
	ESW	−0.6806	0.8897	0.3519	7.68
Impulse Recovery	ESMC	−0.7885	0.0990	0.1402	-
	ESW	−0.7790	0.1011	0.1530	2.33
Step Response	ESMC	0.5244	−0.8207	−0.3734	-
	ESW	0.1463	−0.3102	1.4414	12.01

4.4 Ramp Results

Overall the performance of the evolutionary control algorithms on the ramp was comparable to SMC and switching as can be seen from Figs. 11 (incline) and 12 (decline). We will now look at the incline experiment. While traversing the incline, the original ESMC behaviour outperformed the SMC algorithm by a considerable margin, with a 14% reduction in the velocity variation and settling time, as well as the maximum angle deviation during this phase. The switching algorithms were similar to each other, with $\sim 2\%$ difference in the velocity overshoot and practically identical settling times although the ESW had 40% higher angular overshoot (see Table 4). Interestingly, in this experiment, the ESMC had more optimal performance characteristics any other algorithm. The 5° switching angle of the original algorithm proved detrimental in this scenario as the transient behaviour causes a slower and more jittery response. The ESW algorithm evolved to utilize a slightly higher switching angle and thus was not impaired by a switch in the control type during its ascent.

In the second experiment, the one featuring the decline, (Fig. 12) the responses vary more drastically. There is a striking difference between the evolutionary algorithms and their analytical counterparts. The ESMC algorithm has a settling time 50% less than the SMC, with a lower variation in the velocity and a more oscillatory angular deviation as a consequence, a large improvement and demonstration of the flexibility of the control surface obtained from a simple step response.

The ESW algorithm has a much smaller overshoot and settling time for both metrics compared to the original switching. The transient behaviour between 1.5 and 2.2 s of the original switching algorithm has a significant negative impact on its performance. The presumption is that due to the EA, the ESW algorithm is able to tune the switching angle (12.01° in this case) to eliminate this transience, consequently improving all metrics.

This implies that switching algorithm as a control type has a promising margin of improvement which needs additional work to be quantified. Furthermore, the ESMC response is comparable to the ESW response, demonstrating that certain alternative SMC parameters and switching angle combinations are capable of considerable improvements.

4.5 Variable Inclination Results

For these tests, the same surface developed in the step response simulation is used. Each control mode behaved radically different in the variable surface tests, a reliable and trustworthy judgement and comparison on their performance is not feasible. As can be seen from a ten second test in Fig. 13 each control mode is fundamentally different, though similar characteristics pertaining to the other tests can be seen identified.

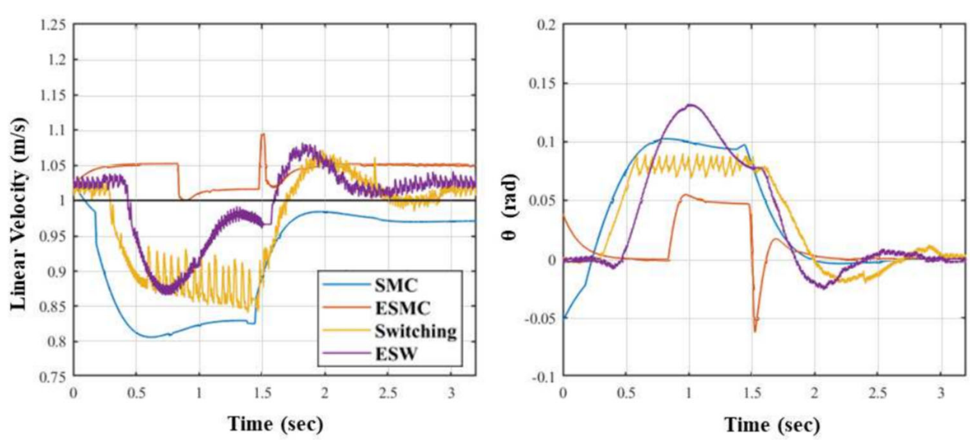


Fig. 11 TWSBR velocity and angular deviation responses while going up the 20° incline

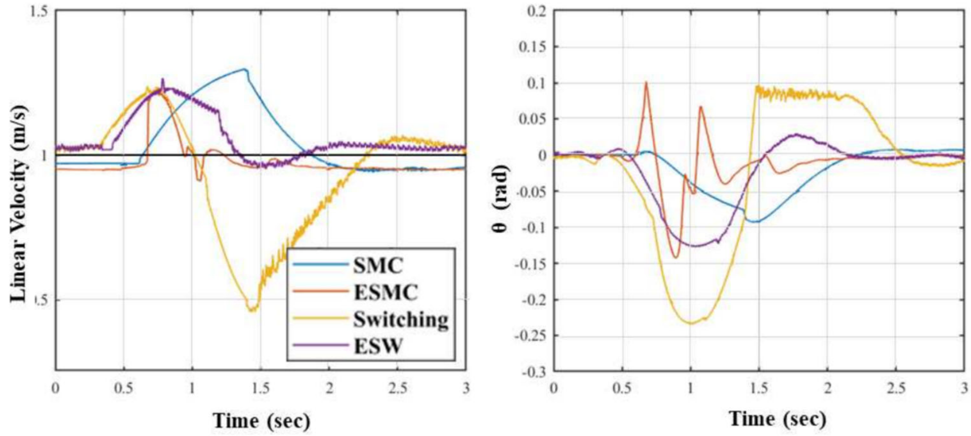


Fig. 12 TWSBR voltage and angular deviation responses while traveling down the 20° decline

The original SMC stays quite close to the velocity target throughout, with small fluctuations as expected. Its angular profile also remains fairly close to the zero point. The ESMC is also able to remain close to the velocity target although at times there are fluctuations larger than the original SMC in the velocity and more so in the angular deviation. This test was run multiple times as each time the response was slightly different. As the above features were consistent in all tests it is reasonable to infer that the ESMC performance is worse at this variable surface than the original SMC.

Both switching algorithms have smaller variability in their velocity throughout compared to the pure SMC algorithms. The original switching algorithm experiences a large velocity drop in a region where there is discord between the surface inclination and control algorithm. Presumably as the switching control alternates between PID and SMC repeatedly, neither control type maintains control long enough to bring the TWSBR to the desired state, thus continuing the cycle until a change in the surface provides the necessary circumstances to escape the transient loop. The ESW algorithm performs just as well as the original switching algorithm, with smaller variability in its velocity and angular deviation profiles and no issues of transience on this surface. For the above reasons, the ESW is considered most competent in such an environment.

The power consumed by the SMC is 188.63 W, the ESMC 177.22 W for the above test, the switching consumed 186.84 W and the ESW 181.79 W. The evolutionary algorithms thus had marginally better power consumption in this test.

4.6 Rough Terrain Results

As expected, the irregularities in the surface negatively affect the TWSBR behaviour (Fig. 14), with all four responses having a considerably more noise. Here the benefits of the analytical SMC over the ESMC are evident as the highly specialised focus of the ESMC becomes a clear disadvantage. Though the settling times are similar, in the velocity profile the original SMC has a 20% steady-state error while the ESMC can eventually settle at the right velocity. The angle profile shows however that the ESMC takes significantly longer to settle. Contrasting this, the two switching algorithms were similar to each other and more apt on the rough terrain, not subject to a steady-state error or excessive oscillations from the surface irregularities as seen in the pure SMC algorithms. Though the switching behaviour is noisy, there is no significant difference between their original step responses, with practically identical velocity overshoots, and no difference in settling times.

Similarly all voltage profiles have more noise as the algorithms adjust to the variable dynamics. The power consumption was highest for the ESMC at 76.00 W, and then 45.89 W, 45.06 W and 45.04 W for the SMC, switching and

Table 4 Comparison of main performance characteristics in ramp experiments

Test	Criterion	SMC	ESMC	Switching	ESW
Ramp Incline	Settling time (s)	2.204	1.674	2.144	1.708
	% Vel. Overshoot	19.33	9.40	15.74	13.43
	% Angle Overshoot	10.3	6.20	8.85	6.61
Ramp Decline	Settling time (s)	1.293	0.768	1.648	1.460
	% Vel. Overshoot	29.9	21.7	53.3	26.1
	% Angle Overshoot	8.98	13.9	10.1	7.52

Best results are in green, worst ones in orange

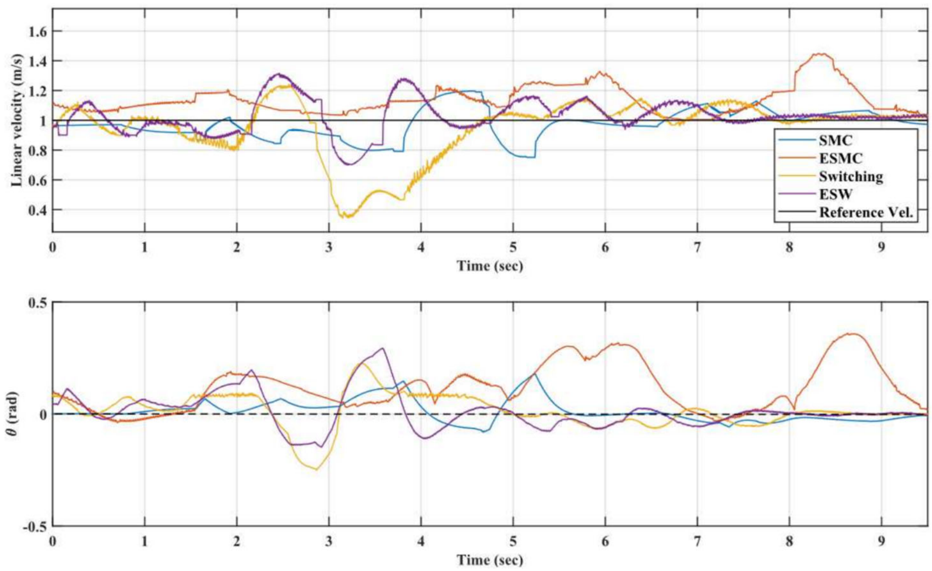


Fig. 13 Velocity and angular deviation profiles of the TWSBR response as it traverses the surface with variable inclination

ESW algorithms respectively, showing that the ESMC does fall short in this task across multiple domains.

4.7 Reduced Friction Results

Angle Recovery The friction tolerance of all four control algorithms was very close as can be seen from Table 5. The best performing algorithm was the ESW in the 55°. The differences however are marginal, can largely be attributed to uncertainties within Vortex, the CoppeliaSim physics engine. The similarity of the friction tolerance values in the algorithms in each test indicates that they are close to their optimal solution in regards to this experiment. This observation serves as an argument in favour of the strengths in both the analytical and evolutionary control algorithms.

Step Response Here the two SMC algorithms perform worse than the switching algorithms. Since the SMC has a 199% velocity overshoot in the step response experiment, it is no surprise that it cannot tolerate much slip before it topples. The ESMC algorithm on the other hand, with a gradual buildup in the velocity step and minimal overshoot, is much better suited to handle reduced friction. It succeeded in tolerating a CoF of 58% before recovery was impossible compared to the minimum 95% of the SMC.

The two switching algorithms are even better and closely matched at a CoF 34% and 35% for the original and ESW algorithms respectively, much superior to the two SMC algorithms. This is likely due to the PID gains keeping the angular deviation consistently low at the start. A side-effect of this is a longer oscillatory response, due to the slower reaction at smaller angles.

5 Discussion on EA Application

These results show that the simheuristic process of obtaining solutions for the TWSBR control parameters can be effective. In every test, the evolutionary algorithms were able to produce values of design variables based on the fitness objective. Notwithstanding clear limitations in the success of this methodology, for example if the robot is inpacted with forces higher than those shown in Fig. 8, the remarkable ability

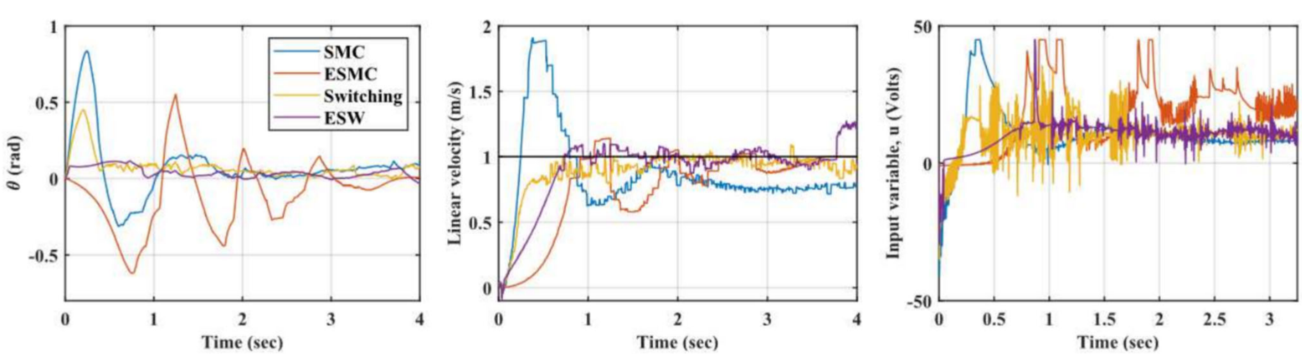


Fig. 14 Velocity, angular deviation, and motor voltage profiles of the four TWSBR control algorithms as it traverses the rough terrain

Table 5 Comparison of minimum CoF before toppling in the reduced friction experiments

Experiment with Reduced Friction	SMC	ESMC	Switching	ESW
55° Inclination	0.68	0.68	0.69	0.65
Step Response	0.95	0.58	0.35	0.34

Best results are in green, worst ones in orange

to influence the direction of this objective solely through the fitness function has the potential to make the EA and simheuristic approach a very valuable tool in this field.

The main challenges faced were the selection of an appropriate generation number and size, meaningful metrics and tuning the fitness function. The tuning of the fitness function does not necessarily produce the expected result, often times the heuristic nature of the algorithm means that the selected metrics may lead a global maximum elsewhere to the assumed location.

The mutation and survivor selection mechanism (offspring creation) had a significant impact on the EA effectiveness. When the mutation factor was large ($\sim$ 0.5% of maximum) the convergence was faster and the population variability remained, but it was too large for global optimum to be reached and maintained on a consistent basis. Essentially a single parameter mutation could cripple a control algorithm. Smaller mutation factors ($\sim$ 0.1% of maximum), though almost twice as slow, were significantly more consistent in their results, an aspect possible with creep mutation. When the mutation factor was zero, parametric changes only occurred based on parent selection, which was completely unproductive as the individuals became very vulnerable to local maxima and completely incapable of escaping them due to the rapidly decreasing genetic diversity.

Two conceivable ways to apply the improvements in practice are to implement additional sensing and active logic to monitor the state of the TWSBR along with the newly evolved control types and switch to the most suitable one depending on what is desired. Another approach would be to try combining all of the above tests into one complex simheuristic environment. It is expected that if the simulation and fitness functions were designed to produce a desired response in multiple different environments, the resultant control algorithms would reflect this.

Figure 15 shows the typical response of the EA population relative to a single performance metric (time upright) in the 55° angle response test with a very low mutation factor. The fitness function was tuned here to select for upright time above all other metrics. The first generations with random parameters rarely stay upright for long, but the ones that immediately accelerate in the direction of the tilt remain upright for a little longer. Thus, the selective pressure leads to individuals slowly increasing the time upright as they accelerate more. Once a stage is reached where they reach the upright position consistently the time upright drastically increases, but not very much as they overshoot and topple in the opposite direction. Eventually the sweet spot is reached, and though few at first, the increased fitness quickly leads to the majority of individuals making it to the end of the simulation (4 s).

This does not correlate directly with other metrics, as an increase in acceleration will decrease the velocity overshoot metric and so on. Thus, although a single metric can be optimised in very few generations and individuals (utilizing a high mutation factor), optimizing all four simultaneously can lead to temporary drops in others, creating a need for larger sample sizes. Once all metrics reach a point of stagnation (fitness plateau) for enough consecutive generations, it is an indication that a global maximum has been reached. Fortunately in this work, the existing analytical algorithms provide a good threshold beyond which any improvements can be assumed to be global maxima.

Our findings align with existing research that leverages Genetic Algorithms (GAs) for optimizing control parameters in self-balancing robots. For instance, Nguyen et al. [53] developed a PID controller for a two-wheeled self-balancing robot, optimizing its parameters using a Genetic Algorithm, which resulted in rapid system stabilization [53]. Similarly, Gao et al. [34] applied a GA to optimize the controller of a single-ball-driven self-balancing robot, enhancing its stability and performance [54]. These studies, along with our findings, demonstrate the efficacy of integrating GAs with traditional control methods to improve the performance and robustness of self-balancing robots.

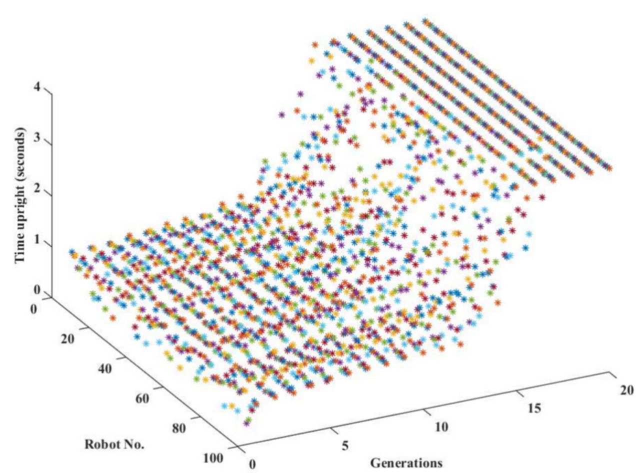


Fig. 15 The time each of the 2500 individuals in an EA population spent upright as the number of generations increased

In assessing the effectiveness of EAs for this control problem, the time needed to develop needs to be considered. As soon as the type of evolutionary algorithm is selected, the structure and implementation is straight forward. There is currently no commonly accepted process to tune the fitness function and EA parameters (generation size and number, state-space, etc.). Since this is a process of trial and error, and where a few hundred individuals must be simulated before any mistake becomes apparent, it is inevitable that it will be a time consuming process.

6 Conclusions and Future Work

This work culminated in the creation of SMC surfaces and switching algorithm combinations that exceed the performance of prior work in most cases, with clear, observable margins. This result provides further proof to the hypothesis that artificial intelligence is instrumental in improving the control of non-linear systems [17]. Certain tests worked better with the EA than others, but importantly, the simheuristic structure was compatible with most aspects of the TWSBR control problem. The evolutionary algorithm was able to produce bigger improvements in the pure SMC control behaviour than in the the switching. This indicates that the SMC is closer to its global optimum and thus only small improvements are possible, further confirmed by the fitness plateaus reached consistently in the final generations.

The evolutionary algorithm was able to produce a clear improvement of the SMC in the 55° recovery, impulse response, variable inclination and some reduced friction tests to varying degrees. In the switching algorithm the evolutionary algorithm produced a better response in the angle recovery tests, the step response, impulse response, all three variable inclination tests and the reduced friction tests. Though the switching response veritably improved in more tests - aided by the tuned PID controller - the margin was small.

The control algorithms were assessed in environments attempting to simulate more realistic and varied characteristics. The largely positive results provide evidence for the versatility of the analytical solutions and the adaptability of the simheuristic approach. The results also serve as an additional and independent confirmation of the effectiveness of the switching control over either PID or SMC on their own.

The main consequence of the results as they relates to practical applications is a baseline assessment of the simheuristic method for tackling control problems in mobile robotics. Further work is required on platforms with more complex movement patterns and constraints.

Since a significant portion of the effort saved in obtaining the parameters is spent tuning and designing the evolutionary algorithm instead, simheuristics need to be carefully considered with regards to the complexity of a problem rather than seen as the immediate answer to all control problems. An important note, seen empirically in the ESMC step-response algorithm, is that the EA is not guaranteed to produce the desired outcome, and may become stuck in local maxima indefinitely. It is also important to note that the solutions that the EA provides may be highly situation dependent, with often poor performance in other situations. With regards to generalization the analytical solution performs better than the EAs. For example the robots that evolved to withstand an impulse often failed at the recovery tests and vice versa.

Though the work has highlighted and strengthened the effectiveness of the novel switching algorithm, further research on the adaptability of the system under more strenuous conditions is necessary. The results seen here consist of only one disturbance mechanism at a time. Thus, future work bringing all of the above results into one coherent control system is certain to be influential in the field. If the results of this study are to be practically useful, additional work must be done on the scalability of the control algorithms and the proposed control algorithm needs to be implemented in a physical robotic platform.

Evolutionary algorithms and simheuritics present an accessible and easily adaptable way of addressing complicated control problems, shifting the burden of analysis and computation away from human involvement. As control problems increase exponentially in scale and complexity the increase in complexity of the EAs required to solve them is linear, providing adequate solutions faster than traditional methods.

Future work will include using the algorithm presented in this work for the control of a physical robot in laboratory tests.

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Author Contributions KDP and NM wrote the code and performed the experiments, MEG and SSA supervised the research and MEG is the corresponding author.

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Declarations

Conflict of interest The authors declare no Conflict of interest with respect to the research, authorship, and/or publication of this article.

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